

CYBERSECURITY THREAT DETECTION AND ANALYSIS SYSTEM

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Objective

The objective of this project is to detect suspicious URLs and analyze cyber threats using Python. The system helps users identify harmful websites and improve cybersecurity awareness.

Problem Statement

Cyber threats such as phishing websites, malicious links, fake login pages and online fraud are increasing rapidly. Users often cannot identify suspicious websites before sharing sensitive information.

Research Findings

Research indicates that phishing attacks are among the most common cybercrimes. Fraudulent websites frequently use misleading keywords and deceptive URLs to trick users.

Methodology

The user enters a URL, the system scans for suspicious keywords, calculates a threat score and displays a threat level based on the analysis.

Tools Used

Python 3, VS Code, GitHub and Windows Operating System.

Implementation

The project uses a keyword-based detection technique to identify suspicious websites and classify them into risk categories.

Results

The developed system successfully identifies suspicious URLs and warns users about potential cyber threats.

Challenges

The system depends on predefined keywords and cannot detect every advanced phishing attack.

Future Scope

Machine Learning, Artificial Intelligence and real-time threat intelligence can be integrated in future versions.

Conclusion

This project demonstrates how Python can be used to create a simple cybersecurity threat detection system.

Python Code

```
suspicious_keywords = ["login", "verify", "bank", "secure",  
"update", "password", "account", "signin", "confirm"] url = input("Enter Website URL: ").lower()  
threat_score = 0 for keyword in suspicious_keywords: if keyword in url: threat_score += 1 if  
threat_score >= 3: print("ALERT: High Risk Website Detected!") elif threat_score >= 1:  
print("Suspicious Website Detected!") else: print("Website Appears Safe")
```